UNIT 8 · STANDARD 6.AT.9

Understand Inequalities and Their Solutions

Lesson 8-5

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can graph the solutions of an inequality on a number line.

Language Objective: I can explain my graph using the words number line, open circle, closed circle, and solution set.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Graph Inequalities | Number line | Open circle | Closed circle | Solution set | Inequality | Boundary point

👆 *Tap any word to see what it means and a picture.*

1. ________ — Show every solution to an inequality on a number line with a boundary point and an arrow.

2. A line with numbers in order used to show values and solutions is a ________.

3. A hollow circle that shows a value is NOT included, used for $<$ or $>$, is an ________.

4. A filled-in circle that shows a value IS included, used for $\leq$ or $\geq$, is a ________.

5. All the values that make an inequality true form the

   ______________________.

6. A math sentence comparing two amounts with <, >, ≤, or ≥ is an

   ______________________.

7. The number where the solution set begins on the number line is the

   ______________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Why did the detective use an open circle instead of a closed circle? What values of p are possible?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Graph Inequalities** — Show every solution to an inequality on a number line with a boundary point and an arrow.
  Representar desigualdades en una recta numérica — Mostrar cada solución de una desigualdad en una recta numérica con un punto límite y una flecha.
- [ ] **Number line** — A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right.
  Recta numérica — Una recta donde los números se colocan en orden; los números son menores a la izquierda y mayores a la derecha.
- [ ] **Open circle** — An empty circle showing a number is NOT included.
  Círculo abierto — Un círculo vacío que muestra que un número NO está incluido.
- [ ] **Closed circle** — A filled-in circle showing a number IS included.
  Círculo cerrado — Un círculo relleno que muestra que un número SÍ está incluido.
- [ ] **Solution set** — All the numbers that make the inequality true.
  Conjunto solución — Todos los números que hacen verdadera la desigualdad.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The detective used an open circle because ____ means 12 is ____ included, so possible values are numbers ____ than 12.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Because 2:00 itself counts, I use a closed circle at 2.

Evidence: Students choose a closed circle at 2 (because 'no earlier than' includes 2) and shade to the right since later times satisfy $t \geq 2$.

Reasoning: This evidence connects the definition of graph inequalities to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective used an open circle because ____ means 12 is ____ included, so possible values are numbers ____ than 12.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.
Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.